

National Institute of Technology, Rourkela
DCCN Lab (CS3072)
6th Semester – 2026 Spring Sem

Evaluation Scheme: -

Day to Day Evaluation: 50 Marks

Viva: 20 Marks

Quiz: 30 Marks

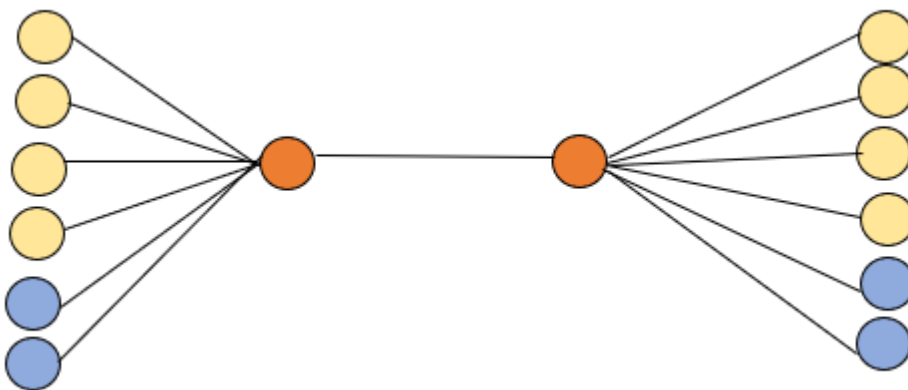
Instructions for the Lab

- A. If a student is absent on the day of evaluation, will be awarded ZERO for that evaluation.
 - B. Turn off your systems before leaving lab. Do not use mobile phone during lab hours.
-

Lab 9

Objective: Performance Comparison of TCP Algorithms Tahoe/Reno: Throughput and Delay

- 1) Write a Tcl script that forms a network consisting of 6 nodes, numbered from 1 to 6. Each of source and destination has bandwidth of 300 Mbps and delay of 20 ms. Set the bottleneck link bandwidth as 500 sec and delay 10ms. Set the queuing discipline to Droptail. Define different colors for different data flows. Send TCP packet from node 1 to node 4 and UDP packet from node 5 to 6. Start the TCP data transmission at 1 sec and UDP at 15 sec. Finish the transmission at 100 sec. Then run nam to view the results. Assume Tahoe/Reno connection between TCP source and sink.



Calculate the following performance metrics using awk script and plot the graph and make the table:

- a) Plot Throughput Graph (Tahoe vs Reno)
- b) Plot End-to-end Delay Graph (Tahoe vs Reno)